PETROLEUM ENGINEERING (PEGN)

PEGN198. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 1-6 Semester Hr.
(I, II) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once. Prerequisite: none. Variable credit; 1 to 6 credit hours. Repeatable for credit under different titles.

PEGN199. INDEPENDENT STUDY. 1-6 Semester Hr.
(I, II) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit; 1 to 6 credit hours. Repeatable for credit.

PEGN201. PETROLEUM ENGINEERING FUNDAMENTALS. 3.0 Semester Hrs.
This course provides an introduction to the oil and gas industry and the various areas associated with petroleum engineering. Topics covered include exploration, development, drilling, production, stimulation, reservoir management, processing, transportation, engineering ethics and professionalism. This course is required for petroleum engineering majors and is open to those interested in petroleum engineering as a minor, and for any other interested students.
Course Learning Outcomes

- Reflect on the origin and importance of petroleum and natural gas in global society.
- Interpret geologic control and subsurface feature data for petroleum engineering applications.
- Justify the role of the main sub-disciplines (drilling, completion, production, and reservoir) of petroleum engineering and how they relate to each other.
- Analyze issues of health, safety, environment, social responsibility, economics, and sustainability as applied in the oil and gas industry.
- Practice using petroleum engineering language derived from testing data, engineering drawings, specifications, and other technical information.
- Analyze equitable and ethical working conditions for all personnel in the field of petroleum engineering.
- Discriminate among career options in the upstream petroleum engineering lifecycle and how these careers depend on, interact with, and support other downstream, midstream, and upstream aspects.

PEGN251. FLUID MECHANICS. 3.0 Semester Hrs.
Introductory and fundamental course in engineering fluid flow. Properties of fluids and fluid flow, fluid statics, mass and momentum balance, differential equations, dimensional analysis, laminar and turbulent flow in pipes, and two-phase flow. Lecture format with demonstrations and practical problem-solving. May not also receive credit for MEGN351 or CEEN310. Prerequisite: CEEN241 or MNGN318.
Course Learning Outcomes

- Define fluid and recognize the limit of this continuum-scale concept.
- Name and describe fluid properties: density, velocity, acceleration, viscosity, surface tension, wetting.
- Classify fluids based on their rheological behaviors.
- Explain and compute hydrostatic pressure and force.
- Apply correlations to compute properties of fluids and simple flows.
- Describe and apply principles of mass/momentum conservation to solve fluid flows.
- Distinguish laminar and turbulent flows.
- Recognize the value of dimensionless correlations and identify dimensionless variables.

PEGN282. PROFESSIONAL SKILLS 1. 1.0 Semester Hr.
This course is the first in a three-course series designed for petroleum engineering students to develop skills in oral and written communication, professionalism, diversity, and ethics. The course is designed as a discussion-based seminar course and will focus on critical thinking and problem-solving. Assignments will be based on technical and non-technical material relating to earth, energy, and the environment. Students will work individually and in multicultural teams on assignments throughout the semester.
Course Learning Outcomes

- Develop critical thinking skills around topics of earth, energy, and environment.
- Demonstrate critical reading skills in field-specific research and technical reports.
- Work effectively in multicultural teams.
- Analyze issues of health, safety, environment, social responsibility, and sustainability as applied to the oil and gas industry.

PEGN298. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 1-6 Semester Hr.
(I, II) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once. Prerequisite: none. Variable credit; 1 to 6 credit hours. Repeatable for credit under different titles.

PEGN299. INDEPENDENT STUDY. 1-6 Semester Hr.
(I, II) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit; 1 to 6 credit hours. Repeatable for credit.

PEGN299. INDEPENDENT STUDY. 1-6 Semester Hr.

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PEGN305. COMPUTATIONAL METHODS IN PETROLEUM ENGINEERING. 2.0 Semester Hrs.
This course is an introduction to computers and computer programming applied to petroleum engineering. Emphasis will be on learning Visual Basic programming techniques to solve engineering problems. A toolbox of fluid property and numerical techniques will be developed. 2 hours lecture; 2 semester hours. Prerequisite: MATH213.
Course Learning Outcomes

- Describe what programming is and how valuable it is to the petroleum field.
- Recognize and apply variable types and operators and construct error-free statements.
- Construct syntax for various programming building blocks such as decisions and loops.
- Analyze an algorithm, decide on appropriate logic, and write statements corresponding to the selected language.

- Identify why a program is not working and/or solve the errors – including invalid syntax, flawed logic, and possible inappropriate numerical technique.
- Develop programs for storing and analyzing data.
- Create programs to solve problems found in the petroleum field such as data analysis, pressure traverses, and fluid property calculations.

PEGN308. RESERVOIR ROCK PROPERTIES. 0-3 Semester Hr.
Introduction to basic reservoir rock properties and their measurements. Topics covered include: porosity, saturations, volumetric equations, land descriptions, trapping mechanism, pressure and temperature gradient, abnormally pressured reservoirs. Darcy's law for linear horizontal and tilted flow, radial flow for single-phase liquids and gases, multiphase flow (relative permeability). Capillary pressure and formation compressibility are also discussed. Co-requisites: CEEN241, PEGN251, PEGN201. 2 hours lecture; 3 hours lab; 3 semester hours.
Course Learning Outcomes

- Recognize reservoir forming (sandstone and carbonate) rock types, their textures, and pore structures.
- Determine porosity, discuss the factors that affect porosity, and describe the methods of determining values of porosity.
- Describe methods of determining fluid saturation in reservoir rock and show the relationship between fluid saturation and capillary pressure.
- Define permeability and its determinants and measurement methods for determining values of absolute permeability.
- Handle and prepare cores according to standard core handling procedures.
- Discuss effective and relative permeability; reproduce typical relative permeability curves and show the effect of saturation history on relative permeability.
- Demonstrate the techniques of averaging porosity, fluid saturation, permeability, and reservoir pressure data.
- Reproduce the Darcy equation in differential form, explain its meaning, and integrate the equation for typical reservoir systems.
- Explain boundary tension and wettability and their effect on capillary pressure. Convert laboratory capillary pressure values to reservoir conditions
- Define and describe unconventional (oil and gas) resources and geothermal energy

PEGN311. DRILLING ENGINEERING. 3.0 Semester Hrs.
Study of drilling operations, rig equipment and procedures, wellbore construction processes and planning, drilling fluid design, hydraulics, well control, bit selection and drill string design, directional drilling, and completion equipment. Prerequisite: PEGN251 with a grade of C or higher, PEGN315, CEEN241. Co-requisite: PEGN305.
Course Learning Outcomes

- Memorize and explain drilling operational equipment and procedures.
- Calculate standard drilling engineering values.
- Start the construction of the well-planning process.
- Apprise and select drill bits, BHA's, and tubular such as drill pipe and casing sizes.
- Plan and assess wellbore trajectories for directional and horizontal wellbores.
- Predict hydraulic pressure magnitudes and power losses.
- Recognize and perform basic well control procedures.
- Assess drilling fluid types and measure their properties.
- Review and classify various well completion types, equipment, and procedures.

PEGN312. PROPERTIES OF PETROLEUM ENGINEERING FLUIDS. 3.0 Semester Hrs.
(WI) Properties of fluids encountered in petroleum engineering including reservoir, drilling, and completion fluids, and oilfield waters. Phase behavior, density, viscosity, interfacial tension, and composition of oil, gas, and brine systems. Interpreting lab data for engineering applications. Flash calculations with k-values and equation of state. Introduction to fluid properties software. Laboratory experimentation of fluid properties. Prerequisites: PEGN308 (C or better), CHGN209 (C or better). 2 hours lecture; 3 hours lab; 3 semester hours.
Course Learning Outcomes

- Recognize hydrocarbons and non-hydrocarbons that are commonly seen in oil and gas reservoirs and their influences on fluid properties.
- Use phase diagrams of mixtures to calculate the physical properties of gases and liquids and classify reservoir fluids.
- Calculate the behavior of hydrocarbon mixtures in the two-phase region.
- Recognize the different types of reservoir fluids.
- Estimate properties of dry and wet gas reservoirs.
- Describe the physical properties of black oils, conduct laboratory procedures to characterize black oil properties, and recognize correlations that can be used to determine black oil properties.
- Design a laboratory experiment related to black oil properties.
- Separator calculation for black oils and other reservoir fluid types.
- Describe the properties of oilfield waters and their solution chemistry.

PEGN315. SUMMER FIELD SESSION I. 1.0 Semester Hr.
This 8 day course taken after the completion of the sophomore year is designed to introduce the student to oil and gas field and other engineering operations. Engineering design problems are integrated throughout the 8 day session. On-site visits to various oil field operations in the past included the Rocky Mountain region, the U.S. Gulf Coast, California, Alaska, Canada and Europe. Topics covered include drilling, completions, stimulations, surface facilities, production, artificial lift, reservoir, geology and geophysics. Also included are environmental and safety issues as related to the petroleum industry. 3 hours lab; 1 semester hour. Prerequisite: PEGN308 (grade C or better).
Course Learning Outcomes

- Identify and differentiate between types of oil and gas industry companies and employment opportunities.
- Observe and describe petroleum engineering field and office practices including drilling, completions, production, operations, decommissioning, stakeholder engagement and social responsibility through writing and oral communications.
- Describe and compare field and office experiences with colleagues sharing similar experiences.
- Identify oil and gas company ethical and professional responsibilities and their impact on global, economic, and societal contexts including health, safety, environment, and sustainability.

PEGN316. SUMMER FIELD SESSION II. 2.0 Semester Hrs.
This two week course is taken after the completion of the junior year. Emphasis is placed on the multidisciplinary nature of reservoir management. Field trips in the area provide the opportunity to study eolian, fluvial, lacustrine, near shore, and marine depositional systems. These field trips provide the setting for understanding the complexity of

each system in the context of reservoir development and management. Petroleum systems including the source, maturity, and trapping of hydrocarbons are studied in the context of petroleum exploration and development. Geologic methods incorporating both surface and subsurface data are used extensively. 6 hours lab; 2 semester hours. Prerequisite: PEGN315, PEGN419, GEOL308, and GEOL315.

Course Learning Outcomes

- Describe eolian, fluvial, lacustrine, and marine depositional systems and relate them to various drilling, completion, stimulation, production, reservoir, and other petroleum engineering applications
- Explain the complexity that occurs within a single depositional environment and how that affects the overall reservoir development and management.
- Describe structural components of geologic systems and relate them to various drilling, completion, stimulation, production, reservoir, and other petroleum engineering applications.
- Explain how geological systems affect petroleum exploration and development and how these components affect economic decisions.
- Describe how reservoir heterogeneity arises from geologic complexity and may affect field development and management decisions.
- Integrate the knowledge gained to develop a comprehensive plan for acquiring and developing local oil and gas fields.

PEGN340. COOPERATIVE EDUCATION. 0-3 Semester Hr.
(I, II, S) Supervised, full-time, engineering-related employment for a continuous
six-month period (or its equivalent) in which specific educational objectives are achieved. Prerequisite: Second semester sophomore status and a cumulative grade-point average of at least 2.00. 0 to 3 semester hours. Cooperative Education credit does not count toward graduation except under special conditions.

PEGN350. GEOTHERMAL ENERGY. 3.0 Semester Hrs.
Students will learn geothermal energy resources and their utilization based on geoscience and engineering perspectives. Geoscience topics include world distribution of geothermal resources and their classification, heat and mass transfer, geothermal reservoirs, hydrothermal geochemistry, exploration methods, and resource assessment. Engineering topics include thermodynamics of geothermal fluids, power cycles, electricity generation, drilling and well measurements, reservoir-surface engineering, and direct utilization. Economic and environmental considerations and case studies on social acceptance with community are also presented.

PEGN361. COMPLETION ENGINEERING. 3.0 Semester Hrs.
(WI) This class is a continuation from drilling in PEGN311 into completion operations. Topics include casing design, cement planning, completion techniques and equipment, tubing design, wellhead selection, and sand control, and perforation procedures. 3 hours lecture; 3 semester hours. Prerequisite: PEGN311 and CEEN311.

Course Learning Outcomes

- Identify, formulate, and solve well completion engineering problems by applying the principles of fluid mechanics, rock mechanics, and materials science.
- Design and evaluate well completion systems, including casing seat depth, tubing size, and packer selection, using appropriate engineering design principles.
- Calculate and analyze design pressures, burst, collapse, and tensile loads on casing and tubing strings under various operating conditions, ensuring structural integrity and compliance with industry standards.
- Design a primary cementing program to achieve effective zonal isolation and well integrity, incorporating slurry selection, displacement efficiency, and cement evaluation techniques.
- Apply professional and ethical judgment by evaluating health, safety, environmental, and societal impacts of completion technology choices, including sand control, perforation systems, and cementing practices.
- Communicate effectively through preparation and delivery of technical reports, design summaries, and oral presentations on well completion design and analysis.
- Research and apply emerging completion technologies, including expandable liners, advanced perforation systems, and intelligent completion tools.

PEGN382. PROFESSIONAL SKILLS 2. 1.0 Semester Hr.
This course is the second in a three-course series designed for petroleum engineering students to develop skills in oral and written communication, professionalism, diversity and ethics. The course is designed as a discussion based seminar course and will focus on oral and written communication skills. Assignments will be based on technical and non-technical material relating to earth, energy, and the environment. Students will work individually and in multicultural teams on assignments throughout the semester. Prerequisite: PEGN282.

Course Learning Outcomes

- Practice using petroleum engineering language derived from testing data, engineering drawings, specifications, and other technical information.
- Write and present technical reports for engineering and management personnel using petroleum engineering terminology.
- Collaborate with multicultural team members to solve operational problems in petroleum engineering.

PEGN398. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 0-6 Semester Hr.
(I, II) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once. Prerequisite: none. Variable credit; 1 to 6 credit hours. Repeatable for credit under different titles.

PEGN398. SPECIAL TOPICS. 1-6 Semester Hr.

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PEGN399. INDEPENDENT STUDY. 1-6 Semester Hr.
(I, II) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit; 1 to 6 credit hours. Repeatable for credit.

PEGN399. INDEPENDENT STUDY. 1-6 Semester Hr.

PEGN411. MECHANICS OF PETROLEUM PRODUCTION. 3.0 Semester Hrs.
Nodal analysis for pipe and formation deliverability including single and multiphase flow. Natural flow and design of artificial lift methods including gas lift, sucker rod pumps, electrical submersible pumps, and hydraulic pumps. 3 hours lecture; 3 semester hours. Prerequisite: PEGN251, PEGN308 (grade of C or better), PEGN311, and PEGN312.

Course Learning Outcomes

- Create IPR and OPR: Create inflow and outflow performance relationships for oil and gas wells using available information, such as reservoir characteristics, fluid properties, well completion details, wellbore configuration, wellhead specifications, etc.
- Estimate the impacts of various factors on IPR and OPR: Estimate the effects of different factors on the inflow and outflow performance relationships for oil and gas wells, such as reservoir and completion variables, artificial lift, fluid properties, wellbore design, etc.
- Provide production design using Nodal analysis: Analyze, troubleshoot, and optimize oil and gas well production based on available production data using nodal analysis.
- Select and design artificial lift system: Select and design artificial lift systems for oil and gas wells based on expected reservoir and completion conditions.
- Identify flow assurance problems and provide resolutions: Recognize some typical flow assurance problems and surface facilities in the oil and gas industry, and create solutions to solve encountered problems

PEGN414. WELL TESTING AND ANALYSIS. 3.0 Semester Hrs.
Solution to the diffusivity equation. Transient well testing: build-up, drawdown, multi-rate test analysis for oil and gas. Flow tests and well deliverabilities. Type curve analysis. Super position, active and interference tests. Well test design. Prerequisites: MATH225 and PEGN419. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Understand the theoretical basis of well-test analysis.
- Explain the capabilities and limitations of well-test analysis.
- Develop well-test analysis skills - both manual and computerized.
- Estimate reservoir properties from well tests.

PEGN419. INTRODUCTION TO FORMATION EVALUATION AND WELL LOGGING. 3.0 Semester Hrs.
Equivalent with GPGN419,
An introduction to well logging methods, including the relationship between measured properties and reservoir properties. Analysis of log suites for reservoir size and content. Graphical and analytical methods will be developed to allow the student to better visualize the reservoir, its contents, and its potential for production. Use of the computer as a tool to handle data, create graphs and log traces, and make computations of reservoir parameters is required. 3 hours lecture; 3 semester hours. Prerequisites: GEOL315, PHGN200 (grade of C or better).

PEGN422. ECONOMICS AND EVALUATION OF OIL AND GAS PROJECTS. 3.0 Semester Hrs.
Project economics for oil and gas projects under conditions of certainty and uncertainty. Topics include time value of money concepts, discount rate assumptions, measures of project profitability, costs, taxes, expected value concept, decision trees, gambler's ruin, and Monte Carlo simulation techniques. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Understand the key players in the oil and gas industry.
- Create an economic model for oil and gas income and service-producing investments.
- Evaluate the economic model using income or service decision criteria.
- Analyze economic models to identify problem areas.
- Apply the knowledge learned in building and evaluating models into an Authorization for Expenditure presentation (AFE).
- Create and deliver a convincing AFE.
- Understand petroleum reserves classification.
- Understand the basics of taxes (internationally and domestically) and their implications on an economic model.
- Calculate After-Tax Cash Flows and incorporate them into an AFE.
- Evaluate a decline curve and understand its implications on an economic model.
- Understand how price decks are developed and start to predict future prices based on logical assumptions.
- Incorporate risk and uncertainty into a model using software and by-hand calculations.

PEGN423. PETROLEUM RESERVOIR ENGINEERING I. 3.0 Semester Hrs.
Data requirements for reservoir engineering studies. Material balance calculations for normal gas, retrograde gas condensate, solution-gas and gas-cap reservoirs with or without water drive. Primary reservoir performance. Forecasting future recoveries by incremental material balance. 3 hours lecture; 3 semester hours. Prerequisite: PEGN419, PEGN316 and MATH 225 or MATH235.
Course Learning Outcomes

- Differentiate between conventional and unconventional petroleum reservoirs based on geology and identify engineering principles that influence their production behavior.
- Describe the role of rock and fluid properties on mechanisms that drive primary production from petroleum reservoirs.
- Apply scientific and mathematical principles for petroleum systems analysis to reservoir engineering applications for estimation of petroleum reserves.
- Analyze production decline curves to estimate the ultimate recovery of conventional and unconventional petroleum reservoirs.
- Analyze oil-water immiscible displacement processes to estimate oil displacement efficiency.

PEGN424. PETROLEUM RESERVOIR ENGINEERING II. 3.0 Semester Hrs.
Reservoir engineering aspects of supplemental recovery processes. Introduction to liquid-liquid displacement processes, gas-liquid displacement processes, and thermal recovery processes. Introduction to numerical reservoir simula tion, history matching and forecasting. Prerequisite: PEGN423 and PEGN438. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Identify the physics behind supplementary/enhanced oil recovery methods and fluid displacement processes, by describing the effects of fluid and rock properties and their interactions on fluid displacement efficiency.
- Apply mathematical models for the operational design of supplementary oil recovery methods, such as waterflooding, to estimate the microscopic, areal, and vertical displacement efficiencies.
- Describe fundamentals of numerical reservoir simulation technology, by explaining the physical processes represented in governing equations of fluid flow in porous media and describing the components of a numerical reservoir simulator.

- Describe the differences between various numerical discretization techniques, and their strength and limitations, in solving transport equations in porous media.
- Apply numerical reservoir simulation to evaluate flow and displacement processes during supplementary oil recovery operations, and analyze the results obtained from the numerical simulation.
- Describe modeling techniques for unconventional reservoirs to analyze well performance in such reservoirs.
- Classify enhanced oil recovery processes.
- Apply screening methodologies for the selection of enhanced oil recovery methods.

PEGN426. FORMATION DAMAGE AND STIMULATION. 3.0 Semester Hrs.
Skin damage associated with formation damage, well deviation, and perforating. Formation damage mechanisms and causes. Stimulation techniques, including acidizing and fracturing. Calculation of matrix and fracturing rates and pressures. Design of matrix acidizing treatments. Selection/determination of hydraulic fracturing components including rock mechanical properties, in-situ stresses, proppants, fluid types, and diversion. Reservoir considerations in fracture propagation and design. Stimulation diagnostics and their application. Prerequisite: PEGN361 and PEGN411.
Course Learning Outcomes

- Differentiate between various formation damage mechanisms and causes.
- Assess well damage using the concepts of skin and effective wellbore radius for full and partially penetrating wellbores.
- Compare different stimulation methods and their applicability and potential for success in various reservoirs.
- Design a matrix acidizing treatment taking into consideration reservoir conditions, geologic parameters, and pumping components such as acid volumes required, injection rates and pressures, etc.
- Interpret the values of and uses for various hydraulic fracturing variables such as closure pressure, rock mechanical properties, fracture mechanics, etc.
- Conclude from the results of various hydraulic fracturing treatment designs including variations in proppant selection, injection rates, fluid and proppant volumes, etc.

PEGN428. ADVANCED DRILLING ENGINEERING. 3.0 Semester Hrs.
Rotary drilling systems with emphasis on design of drilling programs, directional and horizontal well planning. This elective course is recommended for petroleum engineering majors interested in drilling. Prerequisite: PEGN311, PEGN361. 3 hours lecture; 3 semester hours.

PEGN430. ENVIRONMENTAL LAW AND SUSTAINABILITY. 3.0 Semester Hrs.
(WI) In this course students will be introduced to the fundamental legal principles that are relevant to sustainable engineering project development. General principles of United States (U.S.) environmental regulation pertaining to air quality, water quality, waste management, hazardous substances remediation, regulation of chemical manufacture and distribution, natural resources, and energy will be discussed in parallel with international laws pertaining to environmental protection and human rights. In the context of engineering project design, students will explore legal, societal, and ethical risks, and risk mitigation methodologies. 3 hours lecture; 3 semester hours. Prerequisites: HASS100. Corequisites: HASS200.
Course Learning Outcomes

- Demonstrate knowledge and understanding, verbally and in writing, of domestic and international environmental law and applicable administrative and judicial procedure.
- Write persuasively and effectively through a variety of formal and informal writing exercises and independent research of environmental law, social responsibility, and sustainability issues.
- Apply knowledge of environmental law, social responsibility and sustainability in the design and implementation of a team project that promotes just and sustainable engineering solutions.

PEGN438. PETROLEUM DATA ANALYTICS. 0-3 Semester Hr.
Introduction to elementary probability theory and its applications in engineering and sciences; discrete and continuous probability distributions; parameter estimation; hypothesis testing; linear regression; spatial correlations and geostatistics with emphasis on applications in earth sciences and engineering. 2 hours lecture; 3 hours lab; 3 semester hours. Prerequisite: MATH112 and CSCI128.
Course Learning Outcomes

- Practice using petroleum engineering language derived from reservoir data and data analytics.
- Write technical reports for geoscientists and other engineers using petroleum engineering terminology.
- Collaborate with multicultural team members to solve large problems in subsurface modeling and data analysis.
- Formulate complex data analytics problems related to petroleum engineering.
- Solve complex problems using different approaches and compare them in terms of performance.
- Implement systematic design procedures and determine specifications to meet project requirements.

PEGN439. MULTIDISCIPLINARY PETROLEUM DESIGN. 3.0 Semester Hrs.
Equivalent with GEGN439,GPGN439,
(WI) This is a multi-disciplinary design course that integrates fundamentals and design concepts in geology, geophysics, and petroleum engineering. Students work in integrated teams consisting of students from each of the disciplines. Multiple open-ended design problems in oil and gas exploration and field development, including the development of a prospect in an exploration play and a detailed engineering field study are assigned. Several detailed written and oral presentations are made throughout the semester. Project economics including risk analysis are an integral part of the course. Prerequisites: GEOL308, PEGN316. Co-requisites: PEGN426. 2 hours lecture, 3 hours lab; 3 semester hours.
Course Learning Outcomes

- Develop and apply task-related skills. Task-related skills include analyzing data and information and integrating this information to obtain a solution to an open-ended problem.
- Develop and apply meeting management, brainstorming, and critical team skills. Critical team skills include communication, coordination, leadership, feedback, backup, adaptability, and team orientation.
- Apply a task performance strategy using common processes for multidisciplinary teamwork.
- Design an exploitation strategy for a petroleum field that satisfies relevant technical, professional, and societal constraints.

- Build a project management plan that will be followed, modified, and adapted during the execution of the project to solve an open-ended problem. The project management plan includes a schedule (Gantt charts), budget, tasks, deliverables, resource utilization, and internal milestones.

PEGN440. INTRODUCTION TO THE DIGITAL OILFIELD. 3.0 Semester Hrs.
Capstone course for Petroleum Data Analytics minor. The course starts with an introduction to data analysis and visualization packages. The course then has three projects to include drilling, production, and reservoir data analysis along with data visualization techniques. The student will be required to prepare both oral and written and oral project updates and final results. Prerequisite: PEGN438.
Course Learning Outcomes

- Prepare and analyze data from various petroleum data streams including drilling, completions, stimulation, production, and reservoir management.
- Design petroleum engineering projects that satisfy relevant technical, professional, and societal constraints. These projects will incorporate other associated disciplines and will require Use industry analytical graphical software.
- Apply statistical methods to derive insights into petroleum data sets.
- Interpret petroleum data and derive useful conclusions.
- Independent research (prior knowledge, skills attained in previous courses, original ideas, etc.)
- Build a project business plan. Plan will apply project management skills (schedule, budget, tasks, deliverables, resource utilization, internal milestones, Gantt charts, people, and other available tools)
- Demonstrate professionalism through attendance, demeanor, participation, exhibiting integrity, accepting responsibility, taking initiative, team participation and providing leadership as necessary to ensure project success.
- Create formal and informal communications for individual, team, and industry/company use that document and facilitate progress and enhance the impact of the final design.

PEGN450. ENERGY ENGINEERING. 3.0 Semester Hrs.
Energy Engineering is an overview of energy sources that will be available for use in the 21st century. After discussing the history of energy and its contribution to society, we survey the science and technology of energy, including geothermal energy, fossil energy, solar energy, nuclear energy, wind energy, hydro energy, bio energy, energy and the environment, energy and economics, the hydrogen economy, and energy forecasts. This broad background will give you additional flexibility during your career and help you thrive in an energy industry that is evolving from an industry dominated by fossil fuels to an industry working with many energy sources. Prerequisite: MATH213, PHGN200. 3 hours lecture; 3 semester hours.

PEGN460. FLOW IN PIPE NETWORKS. 3.0 Semester Hrs.
This course will provide an introduction to single and two phase hydraulics phenomena and modeling approaches to calculate pressure/temperature profile, losses along and flow rates along a production system. Furthermore, topics related to pipeline flow control and maintenance such as leak detection, damage prevention, integrity and pipe repairs will be covered. Finally, Federal Pipeline Safety Regulations and Health, Safety, and the Environment (HSE) regulations for the transportation of gas and hazardous liquids by pipeline will be discussed. In addition, this course will provide an introduction in transient theoretical modeling and design applications. OLGA transient multiphase flow simulator will be introduced and used to complete homework and final project. Industrial practices and operational problem related to transient production design will be covered. Prerequisites: PEGN251, CHGN209 , MATH225, and PEGN305. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Estimate local rates, pressure and temperature drops on individual sections of a given pipeline network for single and two-phase flow system under steady state condition.
- Apply required criteria to select required pipe specifications.
- Establish understanding about flow control and pipeline maintenance.
- Understand HSE regulations to transport gas and hazardous liquids.
- Understand and evaluate different transient flow conditions existing in the oil and natural gas industry.
- Use prediction tools to identify and mitigate transient conditions and flow assurance problems for a given production system.
- Provide solutions to eliminate, mitigate or remediate operational problems in a production system.

PEGN461. SURFACE FACILITIES DESIGN AND OPERATION. 3.0 Semester Hrs.
This course will cover surface facilities typically required in the oil and gas industry. The course provides basic operation, design and evaluation of individual equipment such as Control equipment (control valve, pressure/level/flow rate/temperature), Liquid/gas Separators, Flowmeters, Boosting Equipment (pumps, compressors), Heaters, and Storage. Basic principles are described to design and evaluate different midstream processes such as Oil/water treating, Gas/liquid and liquid/liquid separation, Crude oil stabilization, Gas handling facilities, Dehydration, Gas Sweetening, Liquefied Natural Gas (LNG), Gas to Liquids (GTL). Furthermore, potential operation problems and piping and instrumentation diagram/drawing (P&ID) related to this processes will be discussed. Calculation examples and a design project can be given to integrate all acquired knowledge. Furthermore, ASME and API norms related to material selection, equipment selection, operation and maintenance will be discussed. Finally, Health, Safety, and the Environment (HSE) regulations for midstream operations will be discussed. Course objectives include learning how to select and operate different surface equipment required in the oil and natural gas industry, learning how to monitor, troubleshoot and optimize the operation of different surface equipment required in the oil and natural gas industry. Prerequisites: PEGN251, CHGN209, MATH225, PEGN305. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Select different surface equipment, typically required for oil and natural gas production, treatment and transportation, based on expected operating conditions.
- Monitor, troubleshoot and optimize the operation of different surface equipment required in the oil and natural gas industry.
- Design oil, water and gas handling facilities based on the expected operation requirements.

PEGN462. FLOW ASSURANCE. 3.0 Semester Hrs.
This course will cover hydrocarbon production including design and operational issues. Major subjects to be covered include the prediction of hydrates formation, paraffin, asphaltene, scale and sand deposition, and remedial actions. In addition, operational problems such as slugging, emulsions and corrosion will be covered. This course will provide to student's strong background on hydraulic modeling. Prerequisites:

PEGN251, CHGN209, MATH225, and PEGN305. 3 hours lecture; 3 semester hours.

Course Learning Outcomes

- The student outcome are the abilities to understand and evaluate different flow assurance problems existing in the oil and natural gas industry. Furthermore, use prediction tools to identify flow assurance problems for a given production system; provide solutions to eliminate, mitigate or remediate flow assurance problems encountered in production systems.

PEGN463. PETROLEUM MIDSTREAM DESIGN. 3.0 Semester Hrs.
This course will cover the development of an integrated project in the midstream area. In this the students will integrate the knowledge from the midstream classes to solve a given problem with consideration of social responsibility and societal impacts. The objective is to work with several companies from the midstream sector to solve field problems. Furthermore, in this class, we will have some classes to cover more specific subjects with different presenters (i.e. safety, regulations, marketing, environment, new technologies for pipe repairs or inspections, software, process to sell/buy oil, etc), field visits, etc. 3 hours lecture; 3 semester hours. Prerequisite: PEGN460, PEGN461, PEGN462. Co-requisite: PEGN460.

Course Learning Outcomes

PEGN481. PETROLEUM SEMINAR. 2.0 Semester Hrs.
(I) (WI) Written and oral presentations by each student on current energy topics.
This course is designated as a writing intensive course (WI).
Prerequisite: none. 2 hours lecture; 2 semester hours.

Course Learning Outcomes

- Effectively communicate in a variety of formats - both orally and in written form, and with a range of audiences
- Explain, integrate, and evaluate contemporary issues that apply to the oil and gas profession, considering economic, environmental, corporate, social, political, ethical, health, safety, manufacturability, and sustainability issues
- Demonstrate professionalism through attendance, demeanor, and participation
- Effectively function on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

PEGN482. PROFESSIONAL SKILLS 3. 1.0 Semester Hr.
This course is the third in a three-course series designed for petroleum engineering students to develop skills in oral and written communication, professionalism, diversity and ethics. The course is designed as a discussion based seminar course and will focus on oral and written communication skills, professionalism, diversity and ethics. Assignments will be based on technical and non-technical material relating to earth, energy, and the environment. Students will work individually and in multicultural teams on assignments throughout the semester. Prerequisite: PEGN382.

Course Learning Outcomes

- Critique professional and/or technical papers or reports and analyze elements of professional writing, ethics, and engineering soundness.
- Design an ideal team for a given project scenario taking into consideration required functionality for team success, leadership, varying agendas, team member collaboration, inclusive environment, diversity strategies, and Stakeholder requirements.
- Adapt communications to various audiences and stakeholders (e.g., managers, community members, regulators, technicians).

PEGN490. RESERVOIR GEOMECHANICS. 3.0 Semester Hrs.
The course provides an introduction to fundamental rock mechanics and aims to emphasize their role in oil and gas exploration, drilling, completion and production engineering operations. Deformation as a function of stress, elastic moduli, in situ stress, stress magnitude and orientation, pore pressure, strength and fracture gradient, rock characteristic from field data (seismic, logging, drilling, production), integrated wellbore stability analysis, depletion and drilling induced fractures, compaction and associated changes in rock properties, hydraulic fracturing and fracture stability are among the topics to be covered. Pre-requisites: CEEN311. 3 hours lecture; 3 hours lab, 3 semester hours.

PEGN497. SPECIAL SUMMER COURSE. 0-15 Semester Hr.

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PEGN497. SPECIAL SUMMER COURSE. 0-15 Semester Hr.

PEGN498. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 1-6 Semester Hr.
(I, II) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once. Prerequisite: none. Variable credit; 1 to 6 credit hours. Repeatable for credit under different titles.

PEGN498. SPECIAL TOPICS. 1-6 Semester Hr.

PEGN499. INDEPENDENT STUDY. 1-6 Semester Hr.
(I, II) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit; 1 to 6 credit hours. Repeatable for credit.

PEGN499. INDEPENDENT STUDY. 1-6 Semester Hr.

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PEGN501. APPLICATIONS OF NUMERICAL METHODS TO PETROLEUM ENGINEERING. 3.0 Semester Hrs.
The course will solve problems of interest in Petroleum Engineering through the use of spreadsheets on personal computers and structured FORTRAN programming on PCs or mainframes. Numerical techniques will include methods for numerical quadrature, differentiation, interpolation, solution of linear and nonlinear ordinary differential equations, curve fitting and direct or iterative methods for solving simultaneous equations. Prerequisites: PEGN414 and PEGN424. 3 hours lecture; 3 semester hours.

PEGN502. ADVANCED DRILLING FLUIDS. 3.0 Semester Hrs.
The physical properties and purpose of drilling fluids are investigated. Emphasis is placed on drilling fluid design, clay chemistry, testing, and solids control. Prerequisite: PEGN311. 2 hours lecture, 3 hours lab; 3 semester hours.

PEGN503. INTEGRATED EXPLORATION AND DEVELOPMENT. 3.0 Semester Hrs.
(I) Students work alone and in teams to study reservoirs from fluvial-deltaic and valley fill depositional environments. This is a multidisciplinary course that shows students how to characterize and model subsurface reservoir performance by integrating data, methods and concepts from geology, geophysics and petroleum engineering. Activities include field trips, computer modeling, written exercises and oral team presentations. Prerequisite: none. 2 hours lecture, 3 hours lab; 3 semester hours. Offered fall semester, odd years.

PEGN504. INTEGRATED EXPLORATION AND DEVELOPMENT. 3.0 Semester Hrs.
(I) Students work in multidisciplinary teams to study practical problems and case studies in integrated subsurface exploration and development. The course addresses emerging technologies and timely topics with a general focus on carbonate reservoirs. Activities include field trips, 3D computer modeling, written exercises and oral team presentation. Prerequisite: none. 3 hours lecture and seminar; 3 semester hours. Offered fall semester, even years.

PEGN505. HORIZONTAL WELLS: RESERVOIR AND PRODUCTION ASPECTS. 3.0 Semester Hrs.
This course covers the fundamental concepts of horizontal well reservoir and production engineering with special emphasis on the new developments. Each topic covered highlights the concepts that are generic to horizontal wells and draws attention to the pitfalls of applying conventional concepts to horizontal wells without critical evaluation. There is no set prerequisite for the course but basic knowledge on general reservoir engineering concepts is useful. 3 hours lecture; 3 semester hours.

PEGN506. ENHANCED OIL RECOVERY METHODS. 3.0 Semester Hrs.
Enhanced oil recovery (EOR) methods are reviewed from both the qualitative and quantitative standpoint. Recovery mechanisms and design procedures for the various EOR processes are discussed. In addition to lectures, problems on actual field design procedures will be covered. Field case histories will be reviewed. Prerequisite: PEGN424. 3 hours lecture; 3 semester hours.

PEGN508. ADVANCED ROCK PROPERTIES. 3.0 Semester Hrs.
Application of rock mechanics and rock properties to reservoir engineering, well logging, well completion and well stimulation. Topics covered include: capillary pressure, relative permeability, velocity effects on Darcy's Law, elastic/mechanical rock properties, subsidence, reservoir compaction, and sand control. Prerequisites: PEGN423 and PEGN426. 3 hours lecture; 3 semester hours.

PEGN511. ADVANCED THERMODYNAMICS AND PETROLEUM FLUIDS PHASE BEHAVIOR. 3.0 Semester Hrs.
Essentials of thermodynamics for understanding the phase behavior of petroleum fluids such as natural gas and oil. Modeling of phase behavior of single and multi-component systems with equations of states with a brief introduction to PVT laboratory studies, commercial PVT software, asphaltenes, gas hydrates, mineral deposition, and statistical thermodynamics. 3 hours lecture; 3 semester hours. Prerequisites: PEGN312, PEGN305.
Course Learning Outcomes

- Identify the fundamental properties of reservoir rocks and their impact on reservoir behavior, including porosity, saturations, and formation compressibility, for comprehensive understanding in petroleum engineering applications.

PEGN512. ADVANCED GAS ENGINEERING. 3.0 Semester Hrs.
The physical properties and phase behavior of gas and gas condensates will be discussed. Flow through tubing and pipelines as well as through porous media is covered. Reserve calculations for normally pressured, abnormally pressured and water drive reservoirs are presented. Both stabilized and isochronal deliverability testing of gas wells will be illustrated. Prerequisite: PEGN423. 3 hours lecture; 3 semester hours.

PEGN513. RESERVOIR SIMULATION I. 3.0 Semester Hrs.
The course provides the rudiments of reservoir simulation, which include flow equations, solution methods, and data requirement. Specifically, the course covers: equations of conservation of mass, conservation of momentum, and energy balance; numerical solution of flow in petroleum reservoirs by finite difference (FD) and control volume FD; permeability tensor and directional permeability; non-Darcy flow; convective flow and numerical dispersion; grid orientation problems; introduction to finite element and mixed finite-element methods; introduction to hybrid analytical/numerical solutions; introduction to multi-phase flow models; relative permeability, capillary pressure and wettability issues; linear equation solvers; streamline simulation; and multi-scale simulation concept. Prerequisite: PEGN424 or equivalent, strong reservoir engineering background, and basic computer programming knowledge. 3 credit hours. 3 hours of lecture per week.

PEGN515. ADVANCED RESERVOIR ENGINEERING. 3.0 Semester Hrs.
Reservoir Engineering overview. Predicting hydrocarbon in place; volumetric method, deterministic and probabilistic approaches, material balance, water influx, graphical techniques. Fluid flow in porous media; continuity and diffusivity equations. Well performance; productivity index for vertical, perforated, fractured, restricted, slanted, and horizontal wells, inflow performance relationship under multiphase flow conditions. Combining material balance and well performance equations. Future reservoir performance prediction; Muskat, Tarner, Carter and Tracy methods. Fetkovich decline curves. Reservoir simulation; fundamentals and formulation, streamline simulation, integrated reservoir studies. 3 hours lecture, 3 semester hours.

PEGN517. ADVANCED DRILLING ENGINEERING. 3.0 Semester Hrs.
Drilling Engineering overview. Subjects to be covered include overall drilling organization, contracting, and reporting; basic drilling engineering principles and equipment; drilling fluids, hydraulics, and cuttings transport; drillstring design; drill bits; drilling optimization; fishing operations; well control; pore pressure and fracture gradients, casing points and design; cementing; directional drilling and horizontal drilling. 3 hours lecture, 3 semester hours.

PEGN518. ADVANCED PRODUCTION ENGINEERING. 3.0 Semester Hrs.
This course provides the fundamental concepts in the production area, including Inflow Performance Relationships (IPR), Outflow Performance Relationship (OPR) (or Multiphase Flow in Wells), and Nodal Analysis. It also teaches the current most widely used artificial lift methods, including Gas Lift, ESP, and Sucker Rod Pump systems, and their design and application/challenges for conventional and unconventional reservoirs. Also covered are the introduction of major flow assurance issues, such as liquid loading, terrain and severe slugging, wax, gas hydrate, and emulsion, and their prevention and mitigation, as well as the surface facilities for gas and oil processing. Students will have the opportunities

to write multiphase flow simulators, which are essential in production design and flow assurance problems prediction and management. Some widely used commercial production design software (PIPESIM and OLGA) will also be taught and the students will use them in the production system design.

Course Learning Outcomes

PEGN519. ADVANCED FORMATION EVALUATION. 3.0 Semester Hrs.

A detailed review of wireline well logging and evaluation methods stressing the capability of the measurements to determine normal and special reservoir rock parameters related to reservoir and production problems. Computers for log processing of single and multiple wells. Utilization of well logs and geology in evaluating well performance before, during, and after production of hydrocarbons. The sensitivity of formation evaluation parameters in the volumetric determination of petroleum in reservoirs. Prerequisite: PEGN419. 3 hours lecture; 3 semester hours.

PEGN522. ADVANCED WELL STIMULATION. 3.0 Semester Hrs.

(I) Basic applications of rock mechanics to petroleum engineering problems. Hydraulic fracturing; acid fracturing, fracturing simulators; fracturing diagnostics; sandstone acidizing; sand control, and well bore stability. Different theories of formation failure, measurement of mechanical properties. Review of recent advances and research areas. 3 hours lecture; 3 semester hours.

Course Learning Outcomes

PEGN524. PETROLEUM ECONOMICS AND MANAGEMENT. 3.0 Semester Hrs.

Business applications in the petroleum industry are the central focus. Topics covered are: fundamentals of accounting, oil and gas accounting, strategic planning, oil and gas taxation, oil field deals, negotiations, and the formation of secondary units. The concepts are covered by forming companies that prepare proforma financial statements, make deals, drill for oil and gas, keep accounting records, and negotiate the participation formula for a secondary unit. Prerequisite: PEGN422. 3 hours lecture; 3 semester hours.

PEGN530. ENVIRONMENTAL LAW AND SUSTAINABILITY. 3.0 Semester Hrs.

Equivalent with CEEN492,CEEN592,

(II) In this course students will be introduced to the fundamental legal principles that are relevant to sustainable engineering project development. General principles of United States(U.S.) environmental regulation pertaining to air quality, water quality, waste management, hazardous substances remediation, regulation of chemical manufacture and distribution, natural resources, and energy will be discussed parallel with international laws pertaining to environmental protection and human rights. In the context of engineering project design, students will explore legal, societal, and ethical risks, and risk mitigation methodologies. 3 hours lecture; 3 semester hours.

Course Learning Outcomes

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PEGN540. PETROLEUM DATA ANALYSIS. 3.0 Semester Hrs.

This course will take a detailed look at the opportunities, challenges and specific requirements for petroleum data analytics for the energy industry. It starts with an introduction to data analysis and visualization packages. Three projects are assigned in drilling, production, and reservoir data analysis along with data visualization techniques. The student will be required to prepare both oral and written project updates and final results. Prerequisite: PEGN438 or instructor consent.

Course Learning Outcomes

- Contribute toward the significant technical challenges created by large data environments, including architecture, security, integrity, management, scalability, artificial intelligence topics, and distribution.
- Apply the principles and application of informatics, and the goals of enterprise intelligence as applied to the energy industry.
- Prepare and analyze data from various petroleum data streams including drilling, completions, stimulation, production, and reservoir management.
- Interpret petroleum data and derive useful conclusions.
- Demonstrate professionalism through attendance, demeanor, participation, exhibiting integrity, accepting responsibility, taking initiative, team participation and providing leadership as necessary to ensure project success.
- Create formal and informal communications for individual, team, and industry/company use that document and facilitate progress and enhance the impact of the final design.

PEGN547. PHYSICS, MECHANICS AND PETROPHYSICS OF ROCKS. 3.0 Semester Hrs.

(I) This course will discuss topics in rock physics, rock mechanics and petrphysics as outlined below. The class is a combination of lectures, laboratory sessions, and critical reading and discussion of papers. Topics: Stresses, strains, stiffnesses, rock physics, petrophysics: wettability: shale analysis: seismic & log expression of various formations: diagenesis: formation evaluation. 3 hours lecture; 3 semester hours.

Course Learning Outcomes

- First-order Level Learning Objectives • Gain an introduction to and a working knowledge of the main topics in rock physics • Understand and evaluate technical topics related to rock physics applications • Have insight into basic techniques to evaluate reservoirs • Learn tools to assess reserves, and learn best techniques to use rock physics principles
- Second-order Learning Objectives • identify major & minor rock-forming minerals • evaluate or recall elastic properties of major rock-forming minerals • classify mineral constitutents as load-bearing or pore-filling • compute modulus of a dry rock frame constructed with major minerals • know isotropic and other (Major) symmetries • predict modulus changes in fluid and frame with stress • predict modulus changes with cementation • evaluate / defend role of porosity, cementation and diagenesis on elastic properties • evaluate and appraise elastic modulus of frame with geological and well log information • explain differences between static and dynamic stresses, strains and moduli • classify lithological texture to expected acoustic anisotropy • compute elastic bounds: Voigt, Reuss, Hashin-Shtrikman, modified H-S • compute Empirical velocity models

PEGN551. PETROLEUM DATA ANALYTICS - FUNDAMENTALS. 3.0 Semester Hrs.

Introduction to advanced data analytics in the Digital Oilfield. Comprehensive overview of the fundamental building blocks of the digital oilfield from the convergence of operational technology (field instrumentation and control systems) with corporate information technology infrastructure. An understanding of the data foundation for a typical oil and gas exploration and production company and the challenges of Big Data to oilfield operations (volumes, variety, velocity, and data quality). Prerequisite: DSCI403, DSCI530 or MATH530 or Instructor Approval.

Course Learning Outcomes

- A review of the objectives and results from the digital oilfield since 2000 and a discussion of what is new today (lower for much longer oil prices and emerging digital technologies)
- Convergence of OT (operational technology) and IT (information technology) systems. From sensors and control systems (SCADA), to remote decision support environments, to workflow automation, to process optimization
- Review of often used analytical techniques (regression analysis, neural networks, machine learning, deep learning). Machine Learning overview with Python programming and Jypter Notebook platform
- Review of Business Intelligence (reporting), Data Visualization (dashboards, data story telling) and Artificial Intelligence approaches, the strengths, and weaknesses of each.

PEGN552. PETROLEUM DATA ANALYTICS - APPLICATIONS. 3.0 Semester Hrs.
A continuation of the advanced data analytics in the Digital Oilfield. This capstone course will be to apply learnings from the previous sequence of courses to drilling/completions improvement, production analysis, reservoir management optimization, and unconventional resource development. The course requires the ability of the student to be able to collect, manage, manipulate, analyze, develop insights, and report using both written and oral means those insights using good data visualizations. Prerequisite: PEGN551 or Instructor Consent.
Course Learning Outcomes

- The application of data computational techniques for developing and measuring key performance indications for oil and gas, geothermal, and other drilling and/or completion operations
- • The application of predictive analytics for optimization of oil and gas production processes.
- • The application of data analysis techniques for determining best practice for the management of sub-surface resource development.
- • The application of data analysis techniques for predicting and optimizing the development and production of unconventional resources – shale development, tight gas sands, enhanced oil recovery, and other esoteric resource development.

PEGN577. WORKOVER DESIGN AND PRACTICE. 3.0 Semester Hrs.
Workover Engineering overview. Subjects to be covered include Workover Economics, Completion Types, Workover Design Considerations, Wellbore Cleanout (Fishing), Workover Well Control, Tubing and Workstring Design, SlicklineOperations, Coiled Tubing Operations, Packer Selection, Remedial Cementing Design and Execution, Completion Fluids, Gravel Packing, and Acidizing. 3 hours lecture, 3 semester hours.

PEGN590. RESERVOIR GEOMECHANICS. 3.0 Semester Hrs.
The course provides an introduction to fundamental rock mechanics concepts and aims to emphasize their role in exploration, drilling, completion and production engineering operations. Basic stress and strain concepts, pore pressure, fracture gradient and in situ stress magnitude and orientation determination and how these properties are obtained from the field measurements, mechanisms of deformation in rock, integrated wellbore stability analysis, depletion induced compaction and associated changes in rock properties and formation strength, hydraulic fracturing and fracture stability are among the topics to be covered in this rock course. Naturally fractured formation properties and how they impact the characteristics measured in the laboratory and in field are also included in the curriculum. Several industry speakers are invited as part of the lecture series to bring practical aspects of the fundamentals of geomechanics covered in the classroom. In addition, Petrel, FLAC3D and FRACMAN software practices with associated assignments are offered to integrate field data on problems including in situ stress magnitude and orientations, pore pressure and fracture gradient prediction and rock property determination using laboratory core measurements, logs, seismic, geological data. Problems are assign for students to use the field and laboratory data to obtain static and dynamic moduli, rock failure criteria, wellbore stress concentration and failure, production induced compaction/subsidence and hydraulic fracture mechanics.

PEGN591. SHALE RESERVOIR ENGINEERING. 3.0 Semester Hrs.
Equivalent with PEGN615,
Fundamentals of shale-reservoir engineering and special topics of production from shale reservoirs are covered. The question of what makes shale a producing reservoir is explored. An unconventional understanding of shale-reservoir characterization is emphasized and the pitfalls of conventional measurements and interpretations are discussed. Geological, geomechanical, and engineering aspects of shale reservoirs are explained. Well completions with emphasis on hydraulic fracturing and fractured horizontal wells are discussed from the viewpoint of reservoir engineering. Darcy flow, diffusive flow, and desorption in shale matrix are covered. Contributions of hydraulic and natural fractures are discussed and the stimulated reservoir volume concept is introduced. Interactions of flow between fractures and matrix are explained within the context of dual-porosity modeling. Applications of pressure-transient, rate-transient, decline-curve and transient-productivity analyses are covered. Field examples are studied. 3 hours lecture; 3 semester hours.

PEGN592. GEOMECHANICS FOR UNCONVENTIONAL RESOURCES. 3.0 Semester Hrs.
A wide spectrum of topics related to the challenges and solutions for the exploration, drilling, completion, production and hydraulic fracturing of unconventional resources including gas and oil shale, heavy oil sand and carbonate reservoirs, their seal formations is explored. The students acquire skills in integrating and visualizing multidiscipline data in Petrel (a short tutorial is offered) as well as assignments regarding case studies using field and core datasets. The role of integrating geomechanics data in execution of the exploration, drilling, completion, production, hydraulic fracturing and monitoring of pilots as well as commercial applications in unlocking the unconventional resources are pointed out using examples. Prerequisite: PEGN590. 3 hours lecture; 3 semester hours.

PEGN593. ADVANCED WELL INTEGRITY. 3.0 Semester Hrs.
Fundamentals of wellbore stability, sand production, how to keep wellbore intact is
covered in this course. The stress alterations in near wellbore region and associated consequences in the form of well failures will be covered in detailed theoretically and with examples from deepwater conventional wells and onshore unconventionalwell operations. Assignments will be given to expose the students to the real field data to interpret and evaluate cases to determinepractical solutions to drilling and production related challenges. Fluid pressure and composition sensitivity of various formations will be studied. 3 hours lecture; 3 semester hours.

PEGN594. ADVANCED DIRECTIONAL DRILLING. 3.0 Semester Hrs.
Application of directional control and planning to drilling. Major topics covered include: Review of procedures for the drilling of directional wells. Section and horizontal view preparation. Two and three dimensional directional planning. Collision diagrams. Surveying and trajectory calculations. Surface and down hole equipment. Common rig operating procedures, and horizontal drilling techniques. Prerequisite: PEGN311 or equivalent. 3 hours lecture; 3 semester hours.

PEGN596. ADVANCED WELL CONTROL. 3.0 Semester Hrs.
Principles and procedures of pressure control are taught with the aid of a full-scale drilling simulator. Specifications and design of blowout control equipment for onshore and offshore drilling operations, gaining control of kicks, abnormal pressure detection, well planning for wells containing abnormal pressures, and kick circulation removal methods are taught. Students receive hands-on training with the simulator and its peripheral equipment. Prerequisite: PEGN311. 3 hours lecture; 3 semester hours.

PEGN597. TUBULAR DESIGN. 3.0 Semester Hrs.
Fundamentals of tubulars (casing, tubing, and drill pipe) design applied to drilling. Major topics covered include: Dogleg running loads. Directional hole considerations. Design criteria development. Effects of formation pressures. Stability loads after cementing. Effects of temperature, pressure, mud weights, and cement. Helical bending of tubing. Fishing loads. Micro-annulus problem. Strengths of API tubulars. Abrasive wear while rotating drill pipe. How to design for hydrogen sulfide and fatigue corrosion. Connection selection. Common rig operating procedures. Prerequisites: PEGN311 and PEGN361 or equivalent. 3 hours lecture; 3 semester hours.

PEGN598. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 0-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

PEGN598. SPECIAL TOPICS. 0-6 Semester Hr.

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PEGN598. SPECIAL TOPICS. 0-6 Semester Hr.

PEGN598. SPECIAL TOPICS. 0-6 Semester Hr.

PEGN598LA. SPECIAL TOPICS LAB. 0-6 Semester Hr.

PEGN598. SPECIAL TOPICS. 0-6 Semester Hr.

PEGN599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

PEGN599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

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PEGN601. APPLIED MATHEMATICS OF FLUID FLOW IN POROUS MEDIA. 3.0 Semester Hrs.
This course is intended to expose petroleum-engineering students to the special mathematical techniques used to solve transient flow problems in porous media. Bessel’s equation and functions, Laplace and Fourier transformations, the method of sources and sinks, Green’s functions, and boundary integral techniques are covered. Numerical evaluation of various reservoir engineering solutions, numerical Laplace transformation and inverse transformation are also discussed. 3 hours lecture; 3 semester hours.

PEGN604. INTEGRATED FLOW MODELING. 3.0 Semester Hrs.
Students will study the formulation, development and application of a reservoir flow simulator that includes traditional fluid flow equations and a petrophysical model. The course will discuss properties of porous media within the context of reservoir modeling, and present the mathematics needed to understand and apply the simulator. Simulator applications will be interspersed throughout the course. 3 hours lecture; 3 semester hours.

PEGN605. ADVANCED WELL TESTING ANALYSIS. 3.0 Semester Hrs.
Various well testing procedures and interpretation techniques for individual wells or groups of wells. Application of these techniques to field development, analysis of well problems, secondary recovery, and reservoir studies. Productivity, gas well testing, pressure buildup and drawdown, well interference, fractured wells, type curve matching, and shortterm testing. Prerequisite: PEGN426. 3 hours lecture; 3 semester hours.

PEGN608. MULTIPHASE FLUID FLOW IN POROUS MEDIA. 3.0 Semester Hrs.
The factors involved in multiphase fluid flow in porous and fractured media. Physical processes and mathematical models for micro- and macroscopic movement of multiphase fluids in reservoirs. Performance evaluation of various displacement processes in the laboratory as well as in the petroleum field during the secondary and EOR/IOR operations. Prerequisite: PEGN 424, 3 hours lecture; 3 semester hours.

PEGN614. RESERVOIR SIMULATION II. 3.0 Semester Hrs.
The course reviews the rudiments of reservoir simulation and flow equations, solution methods, and data requirement. The course emphasizes multi-phase flow and solution techniques; teaches the difference between conventional reservoir simulation, compositional modeling and multi-porosity modeling; teaches how to construct three-phase relative permeability from water-oil and gas-oil relative permeability data set; the importance of capillary pressure measurements and wetability issues; discusses the significance of gas diffusion and interphase mass transfer. Finally, the course develops solution

techniques to include time tested implicit-pressure-explicitsaturation, sequential and fully implicit methods. Prerequisite: PEGN513 or equivalent, strong reservoir engineering background, and basic computer programming knowledge. 3 credit hours. 3 hours of lecture per week.

PEGN620. NATURALLY FRACTURED RESERVOIRS -- ENGINEERING AND RESERVOIR SIMULATION. 3.0 Semester Hrs.
The subsurface crust of the Earth is highly fractured which impacts mass and energy transport associated with aquifers, petroleum reservoirs, hydrothermal reservoirs, hot-dry-rock enhanced geothermal systems (EGS), and waste disposal in igneous and metamorphic rocks. The course covers reservoir engineering, well testing, and simulation aspects of naturally fractured reservoirs. Specifics include: fractures in subsurface rocks, connectivity and network; fracture flow properties; physical principles underlying reservoir engineering and modeling naturally fractured reservoirs; local and global effects of viscous, capillary, gravity and molecular diffusion flows; dual-porosity/dual-permeability models; multi-scale fracture models; transient testing with non-Darcy flow effects; tracer injection and breakthrough analysis; geomechanics of fractures; compositional model; coal-bed gas model; oil and gas from fractured shale; improved and enhanced oil recovery in naturally fracture reservoirs. Prerequisites: PEGN513 or equivalent, strong reservoir engineering background, and basic computer programming knowledge.
Course Learning Outcomes

- Analyze the significance of mass and energy transport models in subsurface energy resources engineering to inform reservoir performance assessments.
- Apply advance engineering strategies to address complex reservoir system challenges, utilizing critical thinking to conduct and communicate research effectively.
- Develop and implement numerical algorithms for solving fluid flow and heat transport equations in fractured rock environments, demonstrating proficiency with data analysis tools.
- Synthesize computational methods to create solutions for non-linear, real-world mass and heat transport problems, enhancing AI-driven models.
- Design practical solutions to real-world engineering problems by integrating coursework, software, and communication skills.

PEGN624. COMPOSITIONAL MODELING - APPLICATION TO ENHANCED OIL RECOVERY. 3.0 Semester Hrs.
Efficient production of rich and volatile oils as well as enhanced oil recovery by gas injection (lean and rich natural gas, CO2, N2, air, and steam) is of great interest in the light of greater demand for hydrocarbons and the need for CO2 sequestration. This course is intended to provide technical support for engineers dealing with such issues. The course begins with a review of the primary and secondary recovery methods, and will analyze the latest worldwide enhanced oil recovery production statistics. This will be followed by presenting a simple and practical solvent flooding model to introduce the student to data preparation and code writing. Next, fundamentals of phase behavior, ternary phase diagram, and the Peng-Robinson equation of state will be presented. Finally, a detailed set of flow and thermodynamic equations for a full-fledged compositional model, using molar balance, equation of motion and the afore-mentioned equation of state, will be developed and solution strategy will be presented. Prerequisite: PEGN513 or equivalent, strong reservoir engineering background, and basic computer programming knowledge. 3 hours lecture; 3 semester hours.

PEGN660. CARBONATE RESERVOIRS - EXPLORATION TO PRODUCTION. 3.0 Semester Hrs.
Equivalent with GEOL660,
(II) This course will include keynote lectures and seminars on the reservoir characterization of carbonate rocks, including geologic description, petrophysics and production engineering. Course will focus on the integration of geology, rock physics, and engineering to improve reservoir performance. Application of reservoir concepts in hands-on exercises, that include a reflection seismic, well log, and core data. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- 1) working knowledge of carbonate sedimentology and diagenesis its use in construction reservoir geometry and pore system architecture to design effective production strategies; 2) working knowledge of production engineering in carbonate reservoirs with a focus on dual porosity media (matrix and fractures), multi-scale physics and geo-mechanics in fractured carbonate reservoirs to help design reservoir stimulation models; and 3) working knowledge of how to integrate geology, geophysics, petrophysics, and engineering to increase reservoir performance.

PEGN681. PETROLEUM ENGINEERING SEMINAR. 1.0 Semester Hr.
Comprehensive reviews of current petroleum engineering literature, ethics, and selected topics as related to research and professionalism. 0 credit hours, except in students' final semester, in which it will be one credit.

PEGN698. SPECIAL TOPICS IN PETROLEUM ENGINEERING. 0-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

PEGN698. SPECIAL TOPICS. 0-6 Semester Hr.

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PEGN698. SPECIAL TOPICS. 0-6 Semester Hr.

PEGN699. INDEPENDENT STUDY. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

PEGN699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

PEGN699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

PEGN707. GRADUATE THESIS / DISSERTATION RESEARCH CREDIT. 1-15 Semester Hr.
(I, II, S) Research credit hours required for completion of a Masters-level thesis or Doctoral dissertation. Research must be carried out under the direct supervision of the student's faculty advisor. Variable class and semester hours. Repeatable for credit.